Benjamin Luftig

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(415)-827-5276

Programming & Design Skills:

Matlab, Python, R, Java, Arduino, HTML, SolidWorks, Autodesk Fusion,

TECHNICAL EXPERIENCE

MuunBase
Engineering Intern 2026

- Created a small, light, and modular data collection device using consumer electronics and 3D printed materials to analyze civil infrastructure and accessibility
- Produced housings for electronic hardware and software to enable communication between electronics

Separated and Transient Aerodynamics Laboratory, STAL
Undergraduate Researcher 2025-Present

- Researched, designed, and manufactured: vortex chamber, heating elements, and laser temperature sensors to analyze breakdown due to heat
- Gathered and analyzed data on the turbulent effects of a baseball with a wind tunnel and determined corresponding lift and drag coefficients.

Robotics & Mechanisms Laboratory, ROMELA
Undergraduate Researcher 2024-2025

- Designed, built, and tested chassis designs and testing apparatus to create a spring-cable system that minimized interference for testing a monoped robot
- Iterate on a secure fastening system for the power delivery system and accompanying hardware

Bruin Racing
Baja Suspension and Chassis Member 2025-Present

- Modeled and welded the frame and supports for the vehicle, as well as mapped CNC procedure and 3D printed supports for the rear suspension procedure
- Created a mirror apparatus designed with both utility and safety in mind, so in the event of a roll over, the mirrors would collapse away from the driver in a non-destructive manner

Rocket Project at UCLA
Prometheus Ground Systems Lead 2023-2024

Ares Ground Systems Member 2024-2025

- Responsible for planning, development, and creation of the Ground Systems for a hybrid rocket
- Cut costs by 30%, integrated pneumatics system to fill table, while minimizing pressure losses and allowing for modularity, and transitioned the system from NPT to AN fitting s

X1 Robotics
Power Systems Subteam Lead 2024-2025

Chassis Subteam Lead 2025-Present

- Budgeted, and designed power systems for an autonomous quadruped
- Created a standardized testing setup for motors quantifying power-usage, weight, and cost
- Led team to design, build and iterate on the propulsion, actuation, and shell of a spherical robot
- Modeled spherical motion, magnetic calculations, and material simulations

EDUCATION

University of California, Los Angeles
Aerospace Engineering Major Senior (GPA: 3.4)